

How the DR works with OpenShift?

What is the Effective DR Strategy to be followed!

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Disaster Recovery (DR) in OpenShift

Disaster Recovery (DR) in OpenShift ensures that applications, workloads, and cluster configurations can be restored quickly after failures such as hardware crashes, data center outages, or cyberattacks. An **effective DR strategy** depends on Recovery Time Objective (RTO) and Recovery Point Objective (RPO) requirements.

Types of Disaster Recovery Strategies in OpenShift

1. Backup & Restore (Cold DR)

Description: Periodic snapshots of etcd, application data, and persistent volumes are taken and stored off-cluster. Restores are performed manually when needed.

Advantages:

- ✓ Cost-effective (no need for an active standby cluster).
- ✓ Suitable for environments where downtime is acceptable.

Disadvantages:

- ✓ Higher RTO/RPO (longer recovery times).
- ✓ Manual intervention required.

Tools:

- ✓ etcdctl for etcd backup.
- ✓ Velero (or OpenShift API for Data Protection - OADP).
- ✓ rsync or snapshot-based persistent storage backups.

Steps:

1. Take regular snapshots of etcd and store them securely.
2. Backup all Kubernetes resources (oc get all,cm,secret -o yaml).
3. Backup Persistent Volumes using storage snapshots.
4. In a disaster event, deploy a new OpenShift cluster and restore configurations.

To get the OpenShift Free downloads, please visit: <https://assistedcloud.com/free-downloads/>

2. Warm Standby (Pilot Light DR)

Description: A secondary cluster is set up with minimal resources and kept in sync with production using periodic updates.

Advantages:

- ✓ Faster recovery than cold DR.
- ✓ Lower cost than an active-active setup.

Disadvantages:

- ✓ Some data loss may occur depending on sync frequency.
- ✓ Requires additional infrastructure.

Tools:

- ✓ OpenShift GitOps (ArgoCD) for configuration synchronization.
- ✓ Velero/OADP for backup & restore.
- ✓ Storage replication (if supported by storage provider).

Steps:

1. Deploy a minimal OpenShift cluster in another region/data center.
2. Use GitOps to synchronize configurations and applications.
3. Periodically sync Persistent Volumes.
4. In case of failure, scale up resources and restore the latest data snapshot

3. Active-Active (Hot DR)

Description: Multiple OpenShift clusters running simultaneously in different regions, with traffic load balancing between them.

Advantages:

- ✓ Zero downtime (lowest RTO).
- ✓ High availability.

Disadvantages:

- ✓ High infrastructure costs.
- ✓ Requires strong synchronization mechanisms.

Tools:

- ✓ OpenShift Multi-Cluster Management (MCE/ACM).
- ✓ OpenShift Service Mesh or Global Load Balancer (F5, CloudFlare, AWS Route 53).
- ✓ Database replication (MongoDB, PostgreSQL, etc.).
- ✓ RWX storage (CephFS, Portworx) or cloud-native storage.

Steps:

1. Deploy OpenShift clusters across multiple regions.
2. Synchronize configurations using ACM or GitOps.
3. Use a global load balancer to route traffic dynamically.
4. Implement real-time database replication.

Effective OpenShift DR Strategy

The most effective DR strategy depends on business requirements, RTO/RPO, and budget.

Step 1: Define DR Objectives

- ✓ **RTO (Recovery Time Objective):** How quickly the cluster should recover after failure.
- ✓ **RPO (Recovery Point Objective):** Maximum acceptable data loss in case of failure.

S.No	Requirement	Recommended DR Approach
1	RTO: < 5 mins, RPO: 0	Active-Active DR
2	RTO: < 2 hours, RPO: ~1 hour	Active-Passive DR
3	RTO: > 12 hours, RPO: ~1 day	Backup & restore

Step 2: Backup Critical Components

1. **etcd Backup (Cluster Configuration)**
ETCDCTL_API=3 etcdctl snapshot save /backup/etcd-snapshot.db
2. **Application Manifests**
oc get all -n <namespace> -o yaml > backup.yaml
3. **Persistent Volume Backups** (Use Velero or Storage Replication)
4. **Database Backups** (MySQL, PostgreSQL, MongoDB, Kafka, etc.)

Step 3: Choose the Right Replication & Failover Mechanism

Storage Replication: OpenShift Data Foundation (ODF), Ceph, Portworx.

Networking DR:

- ✓ DNS-based failover (Route 53, F5 GTM, GSLB).
- ✓ BGP routing with MetalLB/Calico for on-prem solutions.

Step 4: Automate DR Testing & Failover

- ✓ Use Infrastructure-as-Code (IaC): Automate OpenShift deployment using Ansible/Terraform.
- ✓ Regular Disaster Recovery Drills: Test cluster recovery using Velero restores.
- ✓ Monitor & Alerting: Use Prometheus, Grafana, OpenShift Logging to track cluster health.

Note: "Please refer to the Red Hat OpenShift v4 Disaster Recovery documentation listed below: -

<https://www.redhat.com/en/blog/a-guide-to-high-availability/disaster-recovery-for-applications-on-openshift>

https://docs.openshift.com/container-platform/4.16/virt/backup_restore/virt-disaster-recovery.html

Links to some other useful websites are available: -

<https://portworx.com/blog/disaster-recovery-for-red-hat-openshift-virtualization/>

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